

# Spot the Unfair Test — Set A — Answer Key

Level 4 · Ages 10-13 · Answers in bold. Accept any answer that means the same.

---

- ① Kai wants to know which of two kinds of paper towel soaks up more water. He uses a whole sheet of kind A and half a sheet of kind B, dips each in water for 10 seconds, then squeezes out the water and measures it.

**What is the test trying to find out?**

**Whether the kind of paper towel affects how much water it soaks up.**

---

**What is wrong with it?** **The towel size changed too: kind B had half as much paper, so it would hold less whatever kind it was. (Also accept: tested once.)**

---

**How would you fix it?** **Use pieces of exactly the same size of each kind, and repeat each test a few times.**

---

- ② Mia wants to see if music helps plants grow. She puts one plant with music playing in her sunny bedroom, and one plant with no music in a dark kitchen.

**What is the test trying to find out?**

**Whether music affects how much a plant grows.**

---

**What is wrong with it?** **Two things changed at once: the music AND the place (light and temperature). One plant per group is also too few.**

---

**How would you fix it?** **Keep all the plants in the same place and conditions, with only the music different. Use several plants in each group.**

---

- ③ Tom wants to see if the height a ball is dropped from changes how high it bounces. He drops a tennis ball from 50 cm and a basketball from 100 cm.

**What is the test trying to find out?**

**Whether drop height affects how high a ball bounces.**

---

**What is wrong with it?** **He used two different balls, so the type of ball changed as well as the height. (Also accept: tested once.)**

---

**How would you fix it?** **Drop the same ball from each height, test more heights, and repeat each drop.**

---